

**From Person-Specific Dynamics to Population Inference: A Computationally Efficient
Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel
Dynamic Models**

Ivan Jacob Agaloos Pesigan¹, Michael A. Russell^{1, 2}, and Sy-Miin Chow³

¹Edna Bennett Pierce Prevention Research Center, The Pennsylvania State University

²Department of Biobehavioral Health, The Pennsylvania State University

³Department of Human Development and Family Studies, The Pennsylvania State University

Author Note

Ivan Jacob Agaloos Pesigan  <https://orcid.org/0000-0003-4818-8420>; Michael A. Russell  <https://orcid.org/0000-0002-3956-604X>; Sy-Miin Chow  <https://orcid.org/0000-0003-1938-027X>.

This research was supported by the Prevention and Methodology Training Program (PAMT), through a T32 training grant from the National Institute on Drug Abuse (NIDA; T32 DA017629; MPIs: J. Maggs and S. Lanza), as well as by grant UL1TR002014-06 from the National Center for Advancing Translational Sciences (NCATS).

Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer using SLURM for job scheduling (Yoo et al., 2003), GNU Parallel to run the simulations in parallel (Tange, 2021), and Apptainer to ensure a reproducible software stack (Kurtzer et al., 2017, 2021).

Correspondence concerning this article should be addressed to Ivan Jacob Agaloos Pesigan, Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University, 320 Biobehavioral Health Building, University Park, PA 16802 or by email (ijapesigan@psu.edu).

Links

Research Compendium

The data and materials for this study are available on OSF (<https://osf.io/uenr7>) and GitHub (<https://github.com/jeksterslab/manMetaVAR>, <https://jeksterslab.github.io/manMetaVAR/index.html>).

fitVARMxID R Package

Source code and documentation for the `fitVARMxID` R package are available on GitHub (<https://github.com/jeksterslab/fitVARMxID>, <https://jeksterslab.github.io/fitVARMxID/index.html>).

metaDyn R Package

Source code and documentation for the `metaDyn` R package are available on GitHub (<https://github.com/jeksterslab/metaDyn>, <https://jeksterslab.github.io/metaDyn/index.html>).

Single Replication from the Simulation Study

This vignette presents a complete walk-through of one representative Monte Carlo replication, including data generation, estimation for each competing method, and selected output and diagnostics. Convergence diagnostics for a representative BMLVAR (Bayesian multilevel vector autoregressive model) replication, including PSR, ESS, and trace plots, are provided via the `Mplus` output section of the simulation-replication vignette.

<https://jeksterslab.github.io/manMetaVAR/articles/sim-rep.html>

Empirical Illustration: Meta-Analysis of Affect Dynamics

This vignette reproduces the empirical example reported in the manuscript, including data preprocessing, person-specific VAR estimation, second-stage meta-analytic synthesis, and the resulting population-level affect-dynamics summaries.

<https://jeksterslab.github.io/manMetaVAR/articles/empirical.html>

Benchmarking MetaVAR Against an Uncertainty-Uncorrected Two-Stage Approach and Bayesian Multilevel VAR

This vignette reports the benchmarking analysis comparing the computational demands of MetaVAR, the uncertainty-uncorrected two-stage benchmark, and BMLVAR under a common simulation setting.

<https://jeksterslab.github.io/manMetaVAR/articles/benchmark.html>

Containers for Reproducibility

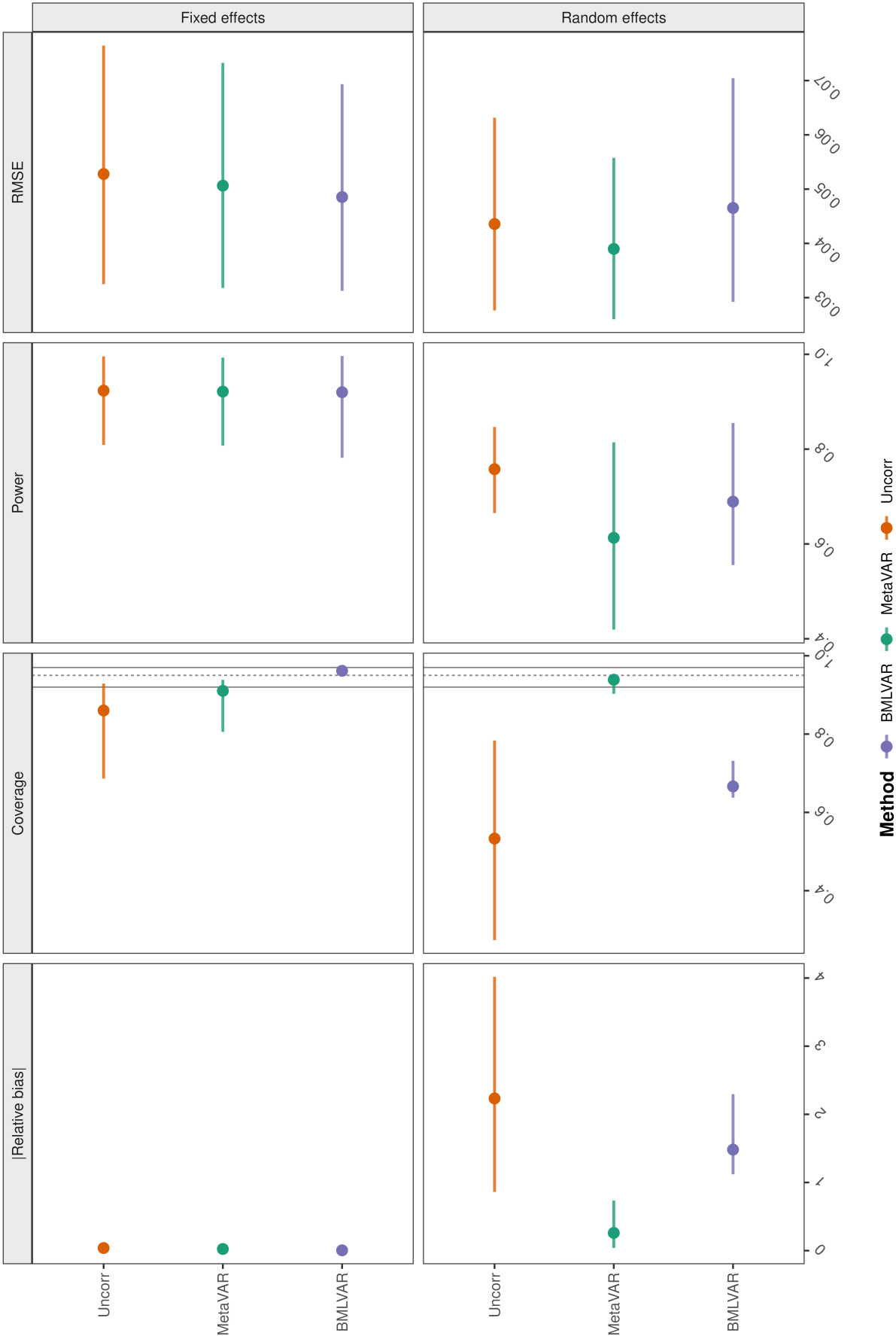
This vignette provides the containerized computing environments and step-by-step instructions needed to reproduce the analyses with the required software stack and dependencies.

<https://jeksterslab.github.io/manMetaVAR/articles/containers.html>

Figure 1
Overall Performance

Overall method performance across simulation design cells

Points are means across design cells; horizontal lines show the min-max range. Coverage panels include nominal .95 and Bradley's liberal .92-.97 band.



Coverage heatmap by parameter, design cell, and method

Figure 1 displays the coverage of the 95% confidence interval for the parameters of the VAR(2) model, comparing three methods: MetaVAR, BMLVAR, and Uncorr. The results are presented in a 3x2 grid of heatmaps, showing coverage for Fixed effects and Random effects across different sample sizes (N = 50, 100, 200, 400) and time series lengths (T = 50, 100, 200, 400). The color scale indicates coverage values from -0.75 (dark red) to 0.00 (white).

The parameters estimated are:

- SP 1
- SP 2
- AR 1->1
- CL 1->2
- CL 2->1
- AR 2->2
- Var(SP 1)
- Cov(SP 1, SP 2)
- Var(SP 2)
- Var(AR 1->1)
- Cov(AR 1->1, CL 1->2)
- Cov(AR 1->1, CL 2->1)
- Cov(AR 1->1, AR 2->2)
- Var(CL 1->2)
- Cov(CL 1->2, CL 2->1)
- Cov(CL 1->2, AR 2->2)
- Var(CL 2->1)
- Cov(CL 2->1, AR 2->2)
- Var(AR 2->2)

The heatmaps show that coverage is generally low (dark red) for most parameters, especially for smaller sample sizes (N = 50) and shorter time series lengths (T = 50). Coverage improves (becomes lighter) as sample size and time series length increase. The Uncorr method shows the lowest coverage, while MetaVAR and BMLVAR show higher coverage, particularly for the random effects.

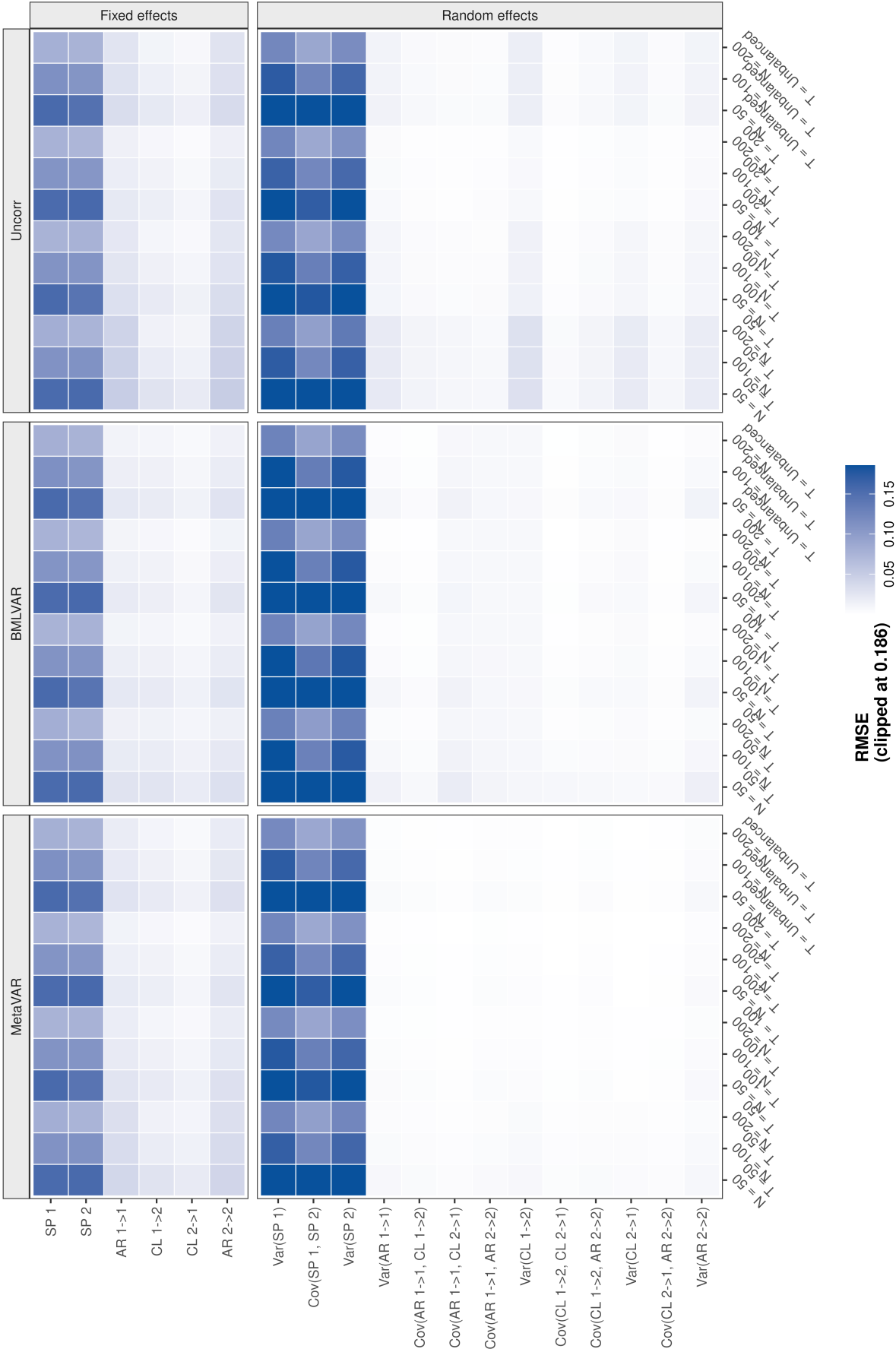
Absolute value of the relative bias heatmap by parameter, design cell, and method

Smaller values are better. The heatmap shows magnitude only, not direction.

Figure 4
RMSE

RMSE heatmap by parameter, design cell, and method

To keep the heatmap readable, RMSE is clipped at the 95th percentile.



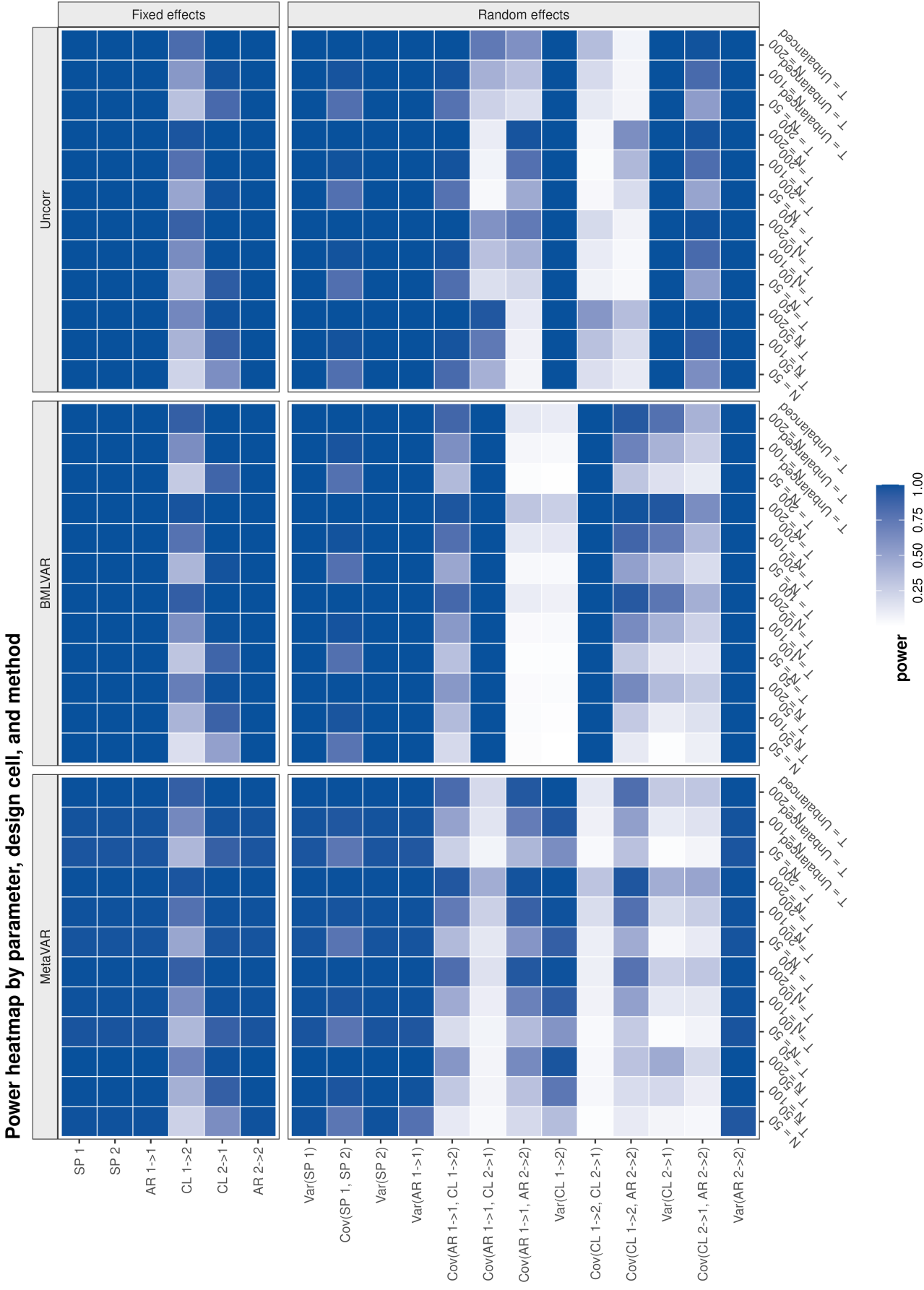


Figure 5
Power